

Project 1 Team Report

Team member	Component(s)	Specific Tasks Completed	Testing/Debugging	GitHub Contributions
Emmanuel F.	Resource Management Reservation Management	• Built Resource class (ID, name, type, availability) • loadResources() to parse resources.txt • displayResources() / displayResourceAvailability() • Built Reservation class • createReservation(), cancelReservation(), displayActiveReservations() • Request validation: resource-exists check, date-format check, duplicate-reservation check	• Confirmed all 20 resources load with correct availability counts • Tested creating a reservation on both available and unavailable resources • Tested invalid resource ID, invalid dates (e.g. 13/40/2026, 02/30/2026), and duplicate student/resource requests • Confirmed auto-assigned reservation IDs don't collide with preloaded data	• Branch: feature/resource-management • Branch: feature/reservation-validation • Commits for Resource/Reservation classes and validation logic • Opened PR, merged to main
Andrew C.	Linked List Implementation Waiting List Management	• Built templated LinkedList<T> (insert, remove, traverse, display) • Built templated Queue<T> (enqueue, dequeue, display) • Built WaitingRequest and WaitingList classes (one FIFO queue per resource) • Built the auto-assignment logic that hands a freed resource to the next waiting student	• Verified insert/remove correctly maintain head/tail pointers • Tested FIFO order with multiple students queued for the same resource • Verified a cancelled reservation correctly triggers auto-assignment to the next waiter • Tested displaying an empty waiting list without crashing	• Branch: feature/linked-list • Branch: feature/waiting-list • Commits for LinkedList<T> and Queue<T> templates, WaitingList class • Opened PR, merged to main

Jacob B.	Cancellation History Complexity Analysis	• Built templated Stack<T> (push, pop, display) • Built CancellationHistory class • undoCancellation() / displayCancellationHistory() • Authored COMPLEXITY_ANALYSIS.md (insertion, removal, waiting-list, undo — all with Big-O)	• Verified cancelled reservations push onto the stack correctly • Tested undo restores the exact reservation and re-marks the resource unavailable • Tested undo with an empty history (no crash) • Cross-checked complexity claims against actual traversal/pointer behavior in the code	• Branch: feature/cancellation-history • Commits for Stack<T> template and CancellationHistory class • Wrote and committed COMPLEXITY_ANALYSIS.md • Opened PR, merged to main
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<https://github.com/emmanuelfakunle22/Group-Project-1-Campus-Resource-Reservation-System/blob/main/README.md>